

Total No. of Questions : 10]

SEAT No. :

P9339

[Total No. of Pages : 3

[6181] - 574

B.E. (Mechanical)

DYNAMICS OF MACHINERY

(2015 Pattern) (Semester - I) (402043)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Draw Neat diagrams wherever necessary.
- 3) Use of scientific calculator is allowed.
- 4) Assume suitable data where ever necessary.
- 5) Figures to the right indicate full marks.

Q1) a) Explain the following terms with respect to the free damped vibrations. **[4]**

- i) Critical damping coefficient
- ii) Damping factor

b) Explain frequency response curves for different damping conditions with significance. **[6]**

OR

Q2) a) Explain following terms: **[4]**

- i) Zero frequency deflection
- ii) Node point

b) An under damped shock absorber is to be designed for a motorcycle of mass 200 kg, such that during a road bump, the damped period of vibration is limited to 2 seconds and the amplitude of vibration should reduce to one-sixteenth in one cycle. Find the spring stiffness and damping coefficient of the shock absorber. **[6]**

Q3) a) Explain with neat diagram mathematical model of a motorbike. **[4]**

- b) Explain following term. **[6]**
- i) Vibration isolation
 - ii) Quality factor
 - iii) Magnification factor

OR

P.T.O.

- Q4)** A periodic torque having a maximum value of 0.65 Nm at a frequency corresponding to 5 rad/sec is impressed upon a flywheel suspended from a wire. The wheel has a moment of inertia of 0.15 kg.m² and a wire is having a diameter of 6 mm and a length of 1.2 m. The modulus of rigidity of the material is equal to 0.8×10^{11} N/m². A viscous dashpot applies damping couple of 0.693 Nm at an angular velocity of 1.5 rad/sec. Calculate: **[10]**
- i) The maximum angular displacement from the rest position
 - ii) The maximum couple applied to dashpot
 - iii) The angle by which the angular displacement lags the torque.

- Q5)** a) Differentiate between static and dynamic balancing. Why there is a need of accurate dynamic balancing of high speed machines? **[6]**
- b) Explain the method of direct and reverse cranks to determine the unbalance forces in radial engines with the help of neat diagrams. **[10]**

OR

- Q6)** a) Explain with neat diagram why single cylinder engine cannot be completely balanced. **[6]**
- b) In a three cylinder radial engine, axes of cylinder are coplanar and angles between neighboring axes are 60°. All cylinders drive same crank of length 220 mm. Each connecting rod is 850 mm long. Reciprocating mass per cylinder is 1.5 kg and the engine runs at 1200 rpm. Find primary and secondary direct and reverse unbalance forces. **[10]**
- Q7)** a) Explain condition monitoring of machines. Explain different techniques for it. **[6]**
- b) Explain the working of FFT Analyzer. **[6]**
- c) It is required to measure the maximum acceleration of a machine, which vibrates violently with the frequency of 700 cycles per min. Accelerometer with negligible damping is attached to it and the indicator travels by 8.2 mm. If the accelerometer weighs 0.5 kg and has a spring rate of 17500 N/m, what is the maximum amplitude and maximum acceleration of the part? **[6]**

OR

Q8) a) What are the different methods of vibration control? Explain any one. [6]

b) Explain working of Magneto-Rheological dampers with neat sketch and application. [6]

c) A vibrometer has a natural frequency of 5 rad/sec and a damping factor of 0.2. An instrument is used to measure a vibrations of a body having a harmonic frequency of 45 rad/sec. The difference between the maximum and minimum reading is 7 mm. Find the amplitude of motion of vibrating body. [6]

Q9) a) Explain in brief various sources of noise and how to control the same. [6]

b) Explain acoustic material & its characteristics. [6]

c) Show that if the sound power is doubled, then the sound power level increases by approximately 3 dB. [4]

OR

Q10) a) Explain following terms with respect to sound : - [6]

i) Anechoic chamber

ii) Reverberation chamber

b) State and explain various types of sound fields? [6]

c) Calculate the total noise, if there are 4 sources of noise having magnitudes 45 dB, 54 dB, 53 dB and 52 dB. What would be effect on total noise, if 45 dB noise is switched off? [4]

